

P5 Archive Browser - Illustrated Workflow Guide

Status: Current workflow companion - v0.31 build 44

Canonical detail: P5 Archive Browser - User Guide

Audience: Archive operators, assistants, and technical reviewers

This guide shows how P5 Archive Browser turns exported inventory into a searchable local catalog, helps an operator identify the right tape, and optionally submits a guarded whole-folder restore to Archiware P5. It is a visual companion to the canonical User Guide, not a replacement for it.

Capability and safety boundary

P5 Archive Browser is local-catalog first. Importing, browsing, searching, Folder Info, notes, Archive Groups, backups, and P5 verification do not modify the P5 server. **Restore Folder from P5 Archive** is the only P5 write path. It is off by default and always requires an explicit confirmation.

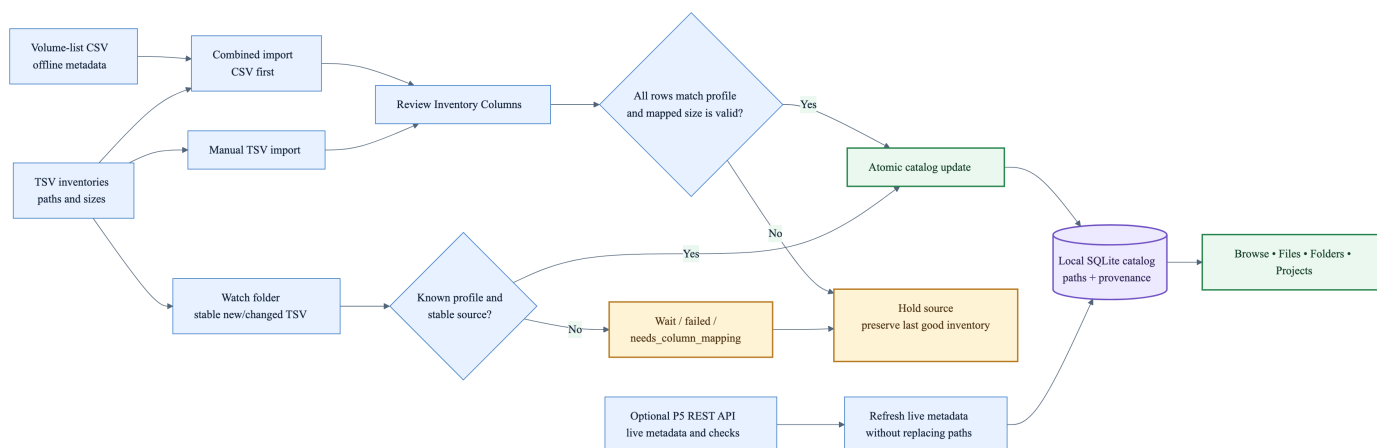
Current behavior:

- Imports supported six- and eight-column TSV inventories only after schema and size validation. A rejected replacement leaves the last good inventory intact.
- Uses TSV data for paths and sizes, CSV for offline volume metadata, and the optional P5 REST API for live metadata and verification.
- Searches Files, Folders, and derived Projects across tapes, then reveals a result in the per-tape browser.
- Restores one whole folder subtree when restore is enabled and P5 is configured. When the destination is readable from this Mac, Browser independently reconciles landed paths and bytes with the local preview.
- Records every restore attempt and preserves durable watch-folder attempts and source provenance.

Current limits:

- Individual-file restore is not supported.
- Automatic watch-folder import handles TSV inventories; the volume-list CSV remains a separate manual import.
- Folder totals come from imported rows and do not yet distinguish directory rows or de-duplicate versioned entries across tapes.
- P5 does not expose a distinct per-job "waiting for tape" state through the endpoint Browser polls. A one-hour Browser timeout means the P5 job may still be running.
- Server identity and routing are still single-server. Named multi-server support is planned.

1. Build a trusted local catalog



Catalog ingest and validation workflow

The recommended first import is **Import > Import Archive Folder (CSV + TSV)...** Browser imports the optional volume-list CSV first so the validated TSV inventories can attach to real P5 labels immediately. Separate CSV and TSV imports remain available.

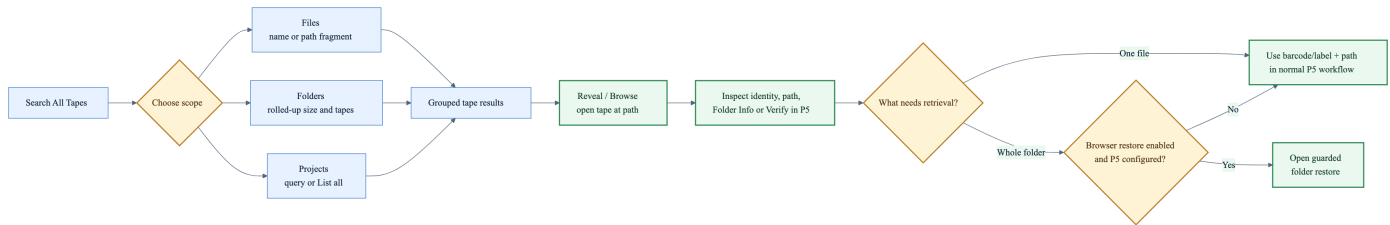
1. Prepare one supported inventory per volume. P5 Archive Export/direct
nsdchat produces six columns; P5 Web GUI Volume Inventory produces eight.
2. Choose a file, a TSV folder, or the combined archive-folder import.
3. Review the detected profile and ordered field mapping before the manual import changes the catalog.
4. Let Browser validate every non-empty row and the mapped nonnegative size.
5. If validation succeeds, Browser atomically replaces that volume's file inventory and schedules search/Projects maintenance.
6. If validation fails, stop and inspect the source. Browser preserves the last successful inventory.

For later exports, **Settings > Watch Folder** can monitor the root plus one generation-folder level. New or changed TSVs wait until stable, network sources are staged locally, and unchanged fingerprints are skipped. An unsupported layout becomes `needs_column_mapping`; background import never guesses.

What each source owns

Source	Provides	Does not prove
TSV inventory	Archived paths, mapped sizes, schema/source provenance	Current tape online state or P5 job state
Volume-list CSV	Offline labels, barcodes, state, type, used size, location	Browsable tape contents
P5 REST API	Live volume metadata, indexes/plans, path verification, restore jobs	That every expected file landed after a restore
Operator fields	Alias, location, notes, Archive Groups	P5-reported metadata

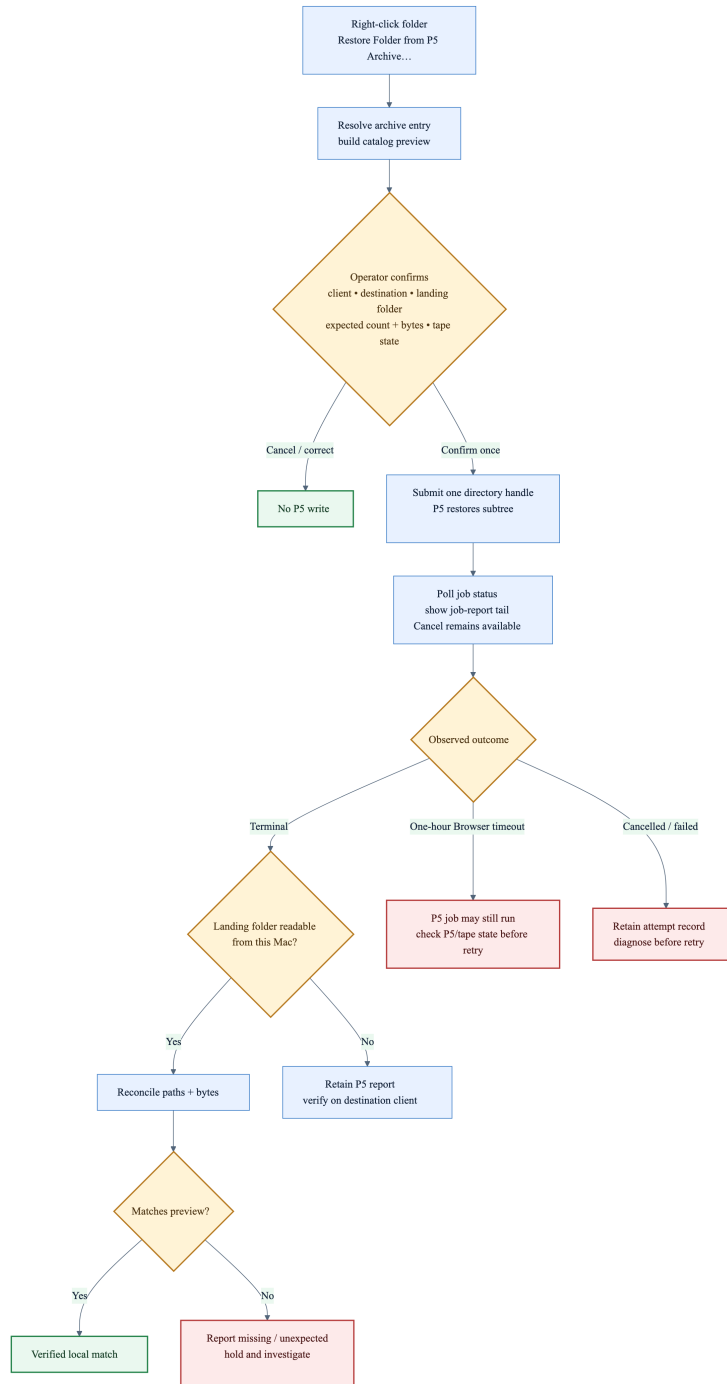
2. Find the tape and inspect the evidence



Search, reveal, and retrieval decision workflow

1. Select **Search All Tapes**.
2. Choose **Files**, **Folders**, or **Projects**.
3. Enter at least three characters for Files or Folders. For Projects, leave the query blank and choose **List** to see all derived projects.
4. Read grouped results by tape. Tape rows expose the best available identity, generation evidence, location, and live online/offline state.
5. Double-click a result or choose **Reveal/Browse** to open the tape at the selected path.
6. Use **Folder Info** for a read-only per-tape and cross-tape roll-up. Use **Verify in P5** on a file when a live server check is useful.
7. Choose the retrieval path:
 - For one file, or when Browser restore is disabled, use the displayed barcode/label and full path in the normal P5 workflow.
 - For a whole folder, use Browser's guarded restore workflow if it is enabled and configured.

3. Restore one whole folder through P5



Guarded whole-folder restore workflow

1. In **Settings > Restore**, enable restore and set default destination values.

The destination path belongs to the selected P5 client; it is not necessarily a path on this Mac.

2. Right-click a folder and choose **Restore Folder from P5 Archive...**

3. Review the confirmation sheet:

- source folder;
- catalog-derived expected count and bytes for the selected tape;

- current online/offline state;
- destination client and path;
- computed landing folder.

4. Confirm once. Browser resolves the P5 archive entry and submits one directory handle, which restores the subtree.

5. Monitor the P5 status and the tail of its job report. Cancel remains available while Browser watches.

6. Interpret the outcome:

- **P5 terminal + local destination readable:** Browser compares paths and bytes and reports match, missing, or unexpected output.

- **P5 terminal + destination not readable here:** retain the P5 report and verify on the destination client.

- **Browser timeout:** the P5 job may still be active, including while waiting for an offline tape. Check P5 directly before retrying.

- **Cancelled or failed:** preserve the attempt record and diagnose before another submission.

The preview describes only the selected tape's catalog rows. A folder archived across multiple tapes can require another tape and P5 may restore more than the preview counted.

Failure and recovery

Signal	Hold action	Safe next check
Unknown, mixed, shifted, malformed, or invalid TSV	Do not replace catalog data	Re-export in a supported profile; review columns again
Watched source is incomplete or unstable	Leave it queued	Let the file stabilize or use Check Now after reconnecting storage
Duplicate sources conflict for one volume	Keep the last good inventory	Remove ambiguity; retry only the intended source
P5 configuration or connection fails	Do not submit restore	Test the connection and confirm the configured server/index
Restore preview does not show the intended landing folder	Cancel at confirmation	Correct destination client/path; reopen the preview
Browser stops watching after one hour	Do not assume job failure	Inspect the P5 job and tape state directly
P5 says complete but local reconciliation differs	Treat as incomplete	Retain the result, inspect missing/extra paths, and verify before retry
Catalog maintenance is required	Back up first	Use Settings > Catalog Data ; reset cannot proceed without its automatic backup

Evidence map

Workflow claim	Public operator reference	Public release evidence
Schema-safe import and preservation of the last good inventory	USER_GUIDE.md	RELEASE_NOTES.md and TESTER_NOTES.md
Stable, deduplicated watch imports and hold states	USER_GUIDE.md	RELEASE_NOTES.md
Cross-tape Files, Folders, and Projects workflows	USER_GUIDE.md	TESTER_NOTES.md
Restore is opt-in and confirmation-gated	USER_GUIDE.md	RELEASE_BLURB_0.30_BUILD_37.md
Restore polling, timeout meaning, and attempt history	USER_GUIDE.md	RELEASE_BLURB_0.31_BUILD_42.md
Independent local path/byte reconciliation	USER_GUIDE.md	TESTER_NOTES.md
Catalog backup and backup-gated reset	USER_GUIDE.md	RELEASE_NOTES.md

The source repository's standalone smoke tests use temporary SQLite fixtures and do not contact a real P5 server or live catalog. This public documentation repository intentionally contains no application source or private test fixtures. Live P5 behavior therefore remains environment- and version-specific.

Planned improvements

These are planned, not shipped:

- Add operator-facing import history and diagnostics for manual and watched

imports. Acceptance gate: every import exposes trigger, source, schema, mapping, counts, result, and a redacted diagnostic export.

- Add named P5 servers and server-scoped tape identity before routing operations

to multiple servers. Acceptance gate: identical labels/IDs on two servers remain distinct and every P5 action routes by explicit provenance.

- Add portable catalog export/import with versioned manifests and verified

rollback. Acceptance gate: a catalog can move without losing inventories, notes, groups, provenance, or compatibility information.

- Consider individual-file restore only after targetPath containment and

traversal behavior are proven. Acceptance gate: selected files retain their intended relative paths and cannot escape the confirmed destination.

- Improve folder totals after directory-row and duplicate/version semantics are

defined and covered by fixtures.

These are future considerations, not commitments for the current pre-release. Use RELEASE_NOTES.md and TESTER_NOTES.md for the current public capability boundary.

Operator acceptance checklist

- [] The running app version matches the procedure being followed.

- [] Import source and detected columns were reviewed.
- [] No held or conflicting source was allowed to replace trusted inventory.
- [] Search result identity, path, and tape were checked before retrieval.
- [] Restore remained disabled unless a whole-folder P5 restore was intended.
- [] Destination client, destination path, computed landing folder, count, and

bytes were reviewed before confirmation.

- [] A timeout was checked in P5 rather than treated as a completed failure.
- [] Local reconciliation or destination-side verification was retained.
- [] Catalog backup exists before repair or reset.

Rebuild the illustrations and PDF

The Mermaid sources and rendered SVG/PNG files live in `docs/workflows/p5-archive-browser/`. The same build also generates `P5_ARCHIVE_BROWSER_ILLUSTRATED_WORKFLOW_GUIDE.pdf`. Regenerate the complete set with:

```
script/build_p5_archive_browser_workflow_guide.sh
```

The build requires Mermaid CLI through `npx` and Python 3 with ReportLab. Set `PDF_PYTHON` when ReportLab is installed in a non-default Python environment.